Dr. James R. Beattie

updated: 12-Oct-2026, website: astro-beattie.com

Personal Information

Nationality:	· Australia · New Zealand
Positions:	· NASA Hubble Fellow & Member at the Institute for Advanced Study
email:	· james.beattie@princeton.edu · jbeattie@cita.utoronto.edu
Online Profiles	· Google Scholar · ResearchGate · OrcID · Twitter
Interests:	· MHD / HD turbulence · plasma astrophysics · multiphase interstellar medium · high-performance computing · supernovae remnants · magnetogenesis · cosmic ray propagation · fluid dynamics · shocks · magnetic dynamos · laboratory astrophysics · statistical modelling · magnetic reconnection · relativistic MHD · neutron star mergers · star formation · interdisciplinary research

Employment

Sep. 2026-2030	**Title:** Hubble Fellow **Institute:** Institute for Advanced Study **Faculty mentor:** Prof. James Stone
Sep. 2026-2030	**Title:** Member **Institute:** Institute for Advanced Study **Faculty mentor:** Prof. James Stone
2023-2027	**Title:** Postdoctoral Research Associate **Institute:** Department of Astrophysical Sciences, Princeton University **Research group:** Prof. Amitava Bhattacharjee
2023-2027	**Title:** CITA Fellow **Institute:** Canadian Institute for Theoretical Astrophysics, University of Toronto **Faculty Mentor:** Prof. Bart Ripperda

Education

2019-2024	**Doctor of Philosophy**, Australian National University, Australia **Specialisation:** Computational / theoretical astrophysics, magnetohydrodynamics. **Thesis**: The statistics of magnetised interstellar turbulence **Advisor:** Christoph Federrath
2018-2019	**Honours (First Class)**, Australian National University, Australia **Major**: Astrophysics **Thesis**: Supersonic Turbulent Molecular Clouds: Filaments and Anisotropies (with University Medal, Chancellor's Commendations, Bok Prize)
2013-2018	**Bachelor of Mathematics**, Queensland University of Technology, Australia **Major**: Applied and Computational Mathematics
2013-2018	**Bachelor of Science**, Queensland University of Technology, Australia **Major:** Physics
2008-2013	**Bachelor of Education**, Queensland University of Technology, Australia **Major**: Biology & Computing

Exchange and Summer Programmes

2022-23	Fulbright Exchange at the University of California, Santa Cruz, United States
2017-18	Cross Institutional Exchange at the University of Queensland, Australia
Winter 2017	Summer Science Programme at The University of Cambridge, United Kingdom
Fall 2015	Exchange Semester at Simon Fraser University, Canada

Scholarships, Awards, Grants & Collaborations

Selected Scholarships & Fellowships

2026 Hubble Fellow, NASA Hubble Fellowship Program
2026 Member, Institute for Advanced Study

2023 CITA Fellow, Canadian Institute for Theoretical Astrophysics
2023 Research Associate, Princeton University
2023 Stanford Science Fellow, Stanford (declined)
2022 Fulbright PhD Fellowship
2020 Joan Duffield Research Supplementary Scholarship
2020 Deakin PhD Scholarship
2020 Dean's Merit (theoretical physics) HDR Supplementary Scholarship
2019 Bok Honours Astrophysics Scholarship
2016, 2017 ANU Summer Research Scholarship

Selected Significant Awards

2025 Top HPC-Enabled Scientific Achievement, HPCwire
2020 Chancellor's Letter of Commendation: 7.0/7.0 Honours GPA
2020 ASA Bok Prize: Best Astronomy Honours Thesis in Australia
2020 Best Student Talk at ANITA, 2020
2019 University Medal (top in graduating science cohort)
2018 Admission to the Dean's List of Students with Excellent Academic Performance
2018 Vice Chancellor's Performance Award
2018 Nominated for 2018 Vice-Chancellor's Awards for Excellence
2016 Vice Chancellor's Performance Award: Best Sessional Teaching in Science & Engineering Faculty
2016 Admission to the Dean's List of Students with Excellent Academic Performance
2015 Admission to the Dean's List of Students with Excellent Academic Performance
2014 Admission to the Dean's List of Students with Excellent Academic Performance

Computing grants awarded (1 core hour $\approx \$0.13$; total ≈ 251 million-core hours)

2026	(CO-PI) SciNet regular call: KHI dynamos and galactic turbulence	1.8e7 core hours
2026	(PI) LRZ regular call: turbulence & dynamos	4e7 core hours
2025	(CO-PI) SciNet regular call: KHI dynamos and galactic turbulence	1.8e7 core hours
2024	(PI) LRZ regular call: KHI dynamos	2e7 core hours
2024	(PI) LRZ large scale call: The world's largest MHD simulation	7e7 core hours
2022	(PI) LRZ large scale call extension	1.5e7 core hours
2021	(PI) LRZ large scale call: The world's largest MHD simulation	7e7 core hours

Collaborations

2026	The Geometry of Flows	Affiliated Scientist

Professional Activities & Organisation Affiliations

Professional Referee Activities

(1 article) Science Advances
(2 articles) Astronomy and Astrophysics
(1 articles) Astronomy and Astrophysics Letters
(1 article) Open Journal of Astrophysics
(2 articles) Astrophysical Journal
(6 articles) Monthly Notices of the Astronomical Society
(1 article) Monthly Notices of the Astronomical Society Letters
(2 article) Journal of Plasma Physics
(1 article) Publications of the Astronomical Society of the Pacific
(first call 2024) LRZ SuperMUC-NG large-scale compute project calls

Other Professional Activities

2023-present Coordinator for the Canadian Institute of Astrophysics Astro-Plasma Group
2022 MSATT program – connecting scientists with high school students
2022 Sustainability Committee, Member, RSAA
2021 Higher Degree Research Education Representative, RSAA
2021 Giving Committee, Member, RSAA
2020 President of the RSAA Student Seminar Committee

Supervisions & Mentoring (13 total)

2026 **Student**: Ish Kaul, grad. Student (co-supervised with S.Peng Oh)
Institute: University of California, Santa Barbara
Project: Unravelling the mass fluxes in the plasma density power spectrum

2026 **Student**: Will Grant, Undergrad. Student
Institute: Amherst College
Project: Stratified, high-plasma beta turbulence for CGM and ICM applications

2025 **Student**: Khush Negandhi (co-supervised w. Bart Ripperda), Undergrad. Student
Institute: University of Toronto
Project: The MHD fluid and nonlinear vector Shrödinger correspondence

2025 **Student**: Jaden Cabral Cordeiro (co-supervised w. Bart Ripperda), Undergrad. Student
Institute: University of Toronto
Project: The transport of ideal invariants in magnetohydrodynamic turbulence

2025 **Student**: Daniel Hagyard (co-supervised w. Bart Ripperda), Undergrad. Student
Institute: University of Toronto
Project: Understanding density gradient statistics for scintillation and laboratory astrophysics

2024-25 **Student**: Isabelle Connor (co-supervised w. Enrico Ramirez-Ruiz), Undergrad. Student
Institute: University of California, Santa Cruz
Project: The universality of supernova-driven turbulence in different galaxies.

2024 **Student**: Max Sokolova (co-supervised w. Bart Ripperda), Undergrad. Student
Institute: University of Toronto
Project: The MHD fluid and nonlinear vector Shrödinger correspondence

2024 **Student**: Juan Miguel L. Padilla (co-supervised w. Bart Ripperda), Undergrad. Student
Institute: University of Toronto
Project: Coherent vortices and eigen modes in MHD turbulence

2024 **Student**: Louis Burnaz (co-supervised w. Bart Ripperda), Undergrad. Student
Institute: École normale supérieure de Lyon
Project: Compression-triggered fast reconnection in relativistic, resistive MHD

2023-24 **Student**: Shashvat Varma (co-supervised w. Bart Ripperda), Undergrad. Student
Institute: University of Toronto
Project: The diffusion-free growth in the small-scale dynamo

2023 **Student**: Sam Lakerdas-Gayle (co-supervised w. Bart Ripperda), Undergrad. Student
Institute: University of Toronto
Project: The secret-life of over-dense regions in magnetised, turbulent clouds

2021 **Student**: Neco Kriel (co-supervised w. Christoph Federrath), Honours Student
Institute: Australian National University
Project: Fundamental scaling relations in the turbulent dynamo.

2021 **Student**: Matthew Sampson (co-supervised w. Mark Krumholz), Honours Student
Institute: Australian National University
Project: Cosmic ray transport in compressible ionised MHD turbulence.

2022 **Student**: Adrian Lehane
Institute: Telopea Park School / Narrabundah College (high school)
Project: Automated phase detection of Venus.

Talks & Meetings (65 total)

Colloquium (1 total)

Mar. 2026 Supernova remnants as engines of non-Kolmogorov galactic turbulence (and why we need to throw away old ideas applied to ISM turbulence), ITC Colloquium, CFA, Harvard University, United States

Collaboration Meetings (2 total)

Feb. 2026 (Invited) Simons Collaboration for Extreme Electrodynamics of Compact Sources, New York, United States
Presented (poster): Can you create a large-scale vertical field from the KHI? Yup.

Feb. 2025 (Invited) Simons Collaboration for Extreme Electrodynamics of Compact Sources, New York, United States
Presented (poster): The Three Phases of the Kelvin Helmholtz Instability Dynamo

Invited (33 total)

Apr. 2027 Cosmology and Particle Physics Seminar, NYU, United States

Mar. 2027 Plasma Seminar, University of Delaware, United States

Nov. 2026 Astrophysics Seminar, Institute of Advanced Studies, United States

Nov. 2026 SFIR meeting, Princeton University, United States

Sep. 2026 The turbulent dynamo across space and time: the chronology of high-k modes
Space and Plasma Seminar, Dartmouth College, United States

Apr. 2026 KIPAC Seminar, Stanford, United States

Mar. 2026 What do the world's largest simulations of cold phase interstellar medium turbulence tell us?
Luncheon CFA talk, Harvard, United States

Mar. 2026	A large-scale, vertical field driven dynamo from the Kelvin-Helmholtz instability: implications for merging compact objects and other thoughts, Blackhole Institute, Harvard, United States
Feb. 2026	Supernova remnants as engines of non-Kolmogorov galactic turbulence ITC Colloquium, Harvard, United States
Feb. 2026	Supernova remnants as engines of non-Kolmogorov galactic turbulence, Astronomy Seminar, Michigan State University, United States
Oct. 2025	Why interstellar turbulence is non-Kolmogorov and how structures fit into turbulence phenomenology, Scintillometry 2025, Montreal, Canada
Sep. 2025	So long Kolmogorov – a new supernova-driven turbulence phenomenology MIST2025 - Cosmic turbulence and Magnetic fields: physics of baryonic matter, France, Corsica
Sep. 2025	Fundamental Results from the World's Largest Simulation of Compressible MHD Turbulence Presented at: AAPPS, Plasma Division, Fukuoka, Japan
Aug. 2025	The chronology of high-k magnetic modes from dynamo to stationary turbulence, Presented at: Frontiers of Relativistic Plasmas Physics in Astrophysics and Laboratory, KITP, Santa Barbara
July. 2025	Magnetized supersonic turbulence with unprecedented dynamical range, Presented at: ASTRONUM, University of Wisconsin, Madison
April. 2025	No phenomenology is safe: surprises from extreme Reynolds number MHD turbulence simulations. Presented at: Caltech tea talk.
Dec. 2024	Supernova-driven turbulence in a multiphase ISM. Presented at: CITA blackboard talk.
Nov. 2024	Unravelling the workings of the compressible turbulent dynamo with $10{,}080^3$ simulations. Presented at: Princeton Plasma Physics Lab, Heliophysics Seminar.
Sep. 2024	The energy cascade II: supernova-driven turbulence. Goutelas, France.
Sep. 2024	The energy cascade I: the world's largest MHD turbulence box. Goutelas, France.
Apr. 2024	Growing a magnetic field in a turbulent plasma. Presented at: CITA Blackboard Seminar.
Apr. 2024	Supersonic magnetised turbulence at extreme Reynolds numbers. Presented at: CITA Theory Seminar
Mar. 2024	Unravelling the fast growth stages of the turbulent dynamo. Presented at: University Maryland Computational and Theory Seminar
Feb. 2024	Numerical resistivity and viscosity. KITP blackboard Talk.
Feb. 2024	A large compressible MHD turbulence simulation with no net magnetic flux. KITP Talk.
Nov. 2023	The most fascinating part of interstellar turbulence: the energy cascade Presented at: TASTY Seminar Series, University of Toronto.
May 2023	The World's Largest Compressible MHD Turbulence Simulation on SuperMUC-NG Presented at: SuperMUC-NG Status and Results Workshop.
Sep. 2022	KIPAC Tea talk: Peta-scale magnetised interstellar medium turbulence simulations. Presented at: SLAC / Stanford University.
Sep. 2022	Magnetised interstellar medium turbulence: dynamics & energetics. Presented at: Susan Clark's research group, Stanford.
Sep. 2022	Astro-coffee: Streaming cosmic rays ion Alfvén velocity statistics. Presented at: Institute for Advanced Study.
Sep. 2022	Bachall lunch discussion: peta-scale simulations & turbulent dynamics. Presented at: Institute for Advanced Study.
Apr. 2022	Streaming cosmic rays ion Alfvén velocity statistics. Presented at: Siang Peng Oh's research group, UC Santa Barbara.
Nov. 2021	Ubiquitous magnetic field fluctuations driven by large-scale supersonic turbulence. Presented at: Star formation and ISM Physics Seminar, Princeton.
Jan. 2021	Ubiquitous magnetic field fluctuations driven by large-scale supersonic turbulence. Presented at: Research School of Astronomy and Astrophysics seminar, ANU.
Aug. 2020	The Anisotropic Density Variance for Highly-Magnetised Molecular Clouds. Presented at: University of Macquarie Colloquium.
Jul. 2020	The Anisotropic Density Variance for Highly-Magnetised Molecular Clouds. Presented at: Astronomical Society of Australia Bok Prize talk.
Jun. 2020	Turbulence at the parsec scale of the Universe. Presented at: Research highlight talk at RSAA full school meeting.
Aug. 2018	The Fractal Geometry of the Supersonic Turbulence in the Interstellar Medium. Presented at: QUT research highlights.
May 2018	The Fractal Geometry of Turbulence. Presented at: QUT Physics Society Meeting.
Nov. 2017	The University of Cambridge and Quantum Mechanics. Presented at: School of Chemistry, Physics and Engineering Seminar, QUT.

Contributed (21 total)

Nov. 2026	Local relaxation and scale-dependent alignment in MHD turbulence Presented at: APS-DPP, Chicago.
Oct. 2026	When Neutron Stars Collide: Unravelling one of the Most Extreme Dynamos in the Universe Presented at: NHFP Symposium, Harvard University.
April 2026	Dynamos. Presented at: CITA blackboard talk.
Nov. 2024	Magnetic dynamos and turbulence: from the cosmos to the lab Presented at: Simons Collaboration for Extreme Electrodynamics of Compact Objects, 2024 meeting, Wisconsin, Madison
Sep. 2024	From Fast Growth to the Saturation of the Intracluster Medium Dynamo Presented at: Galaxy Clusters and Radio Relics II 2024 Workshop, Harvard, Boston
May. 2024	Interstellar medium turbulence and turbulence-regulated star formation theory Presented at: Globular Clusters and Their Tidal Tails: From the Milky Way to the Local Group, Toronto, Canada
May. 2024	The Supersonic Turbulent Dynamo Presented at: HEDLA 2024 Workshop, Tallahassee, Florida
Feb. 2022	Petascale magnetised interstellar medium turbulence simulations Presented at: ANITA 2022 Workshop.
Dec. 2021	Understanding the nature of magnetic field fluctuations driven by large-scale supersonic turbulence. Presented at: Australian Institute of Physics Congress, QUT.
Oct. 2021	Understanding the nature of magnetic field fluctuations driven by large-scale supersonic turbulence. Presented at: Royal Astronomical Society: Galactic magnetic fields meeting.
Feb. 2021	Steps towards anisotropic star formation theory: A multi-shock model for the density variance of anisotropic MHD turbulence. Presented at: ANITA 2021 Workshop.
Dec. 2020	Multi-shock model for the density variance of anisotropic, highly-magnetised ISM turbulence. Presented at: The Magnetic Field Awakens: A new era of star formation.
Nov. 2020	Recent progress on anisotropic, magnetised, supersonic turbulence. Presented at: Mount Stromlo Student Seminars, 2020.
Sep. 2020	Is the Starry Night Turbulent? Presented at: RSAA Feast of Facts.
Feb. 2020	Density, velocity and magnetic structures and correlations in sub-Alfvénic, supersonic turbulence. Accepted* for contributed talk: Magnetic Fields in the Universe 7, Vietnam.
Feb. 2020	Anisotropy in the column density of highly-magnetised supersonic turbulence. Presented at: ANITA 2020 Workshop, UNSW, Canberra.
Dec. 2019	Anisotropic structures in highly-magnetised, observed turbulent clouds. Presented at: Universality: Turbulence across vast scales, Flatiron Inst., New York
Nov. 2019	Reconstructing the 3D Density PDF from the 2D Column Density. Presented at: Cosmic turbulence and magnetic fields : physics of baryonic matter across time and scales in Cargese, France, 2019.
Dec. 2018	Mach number - fractal dimension relation for turbulent, molecular clouds. Poster presented at: AIP Congress 2018, Perth, Australia.
Jan. 2018	The Fractal Geometry of the World's Largest Turbulence Simulation. Presented at: Research School of Astronomy and Astrophysics, ANU.
Jan. 2017	The Analysis of Novel Magnetic Field Configurations in the H-1 NF Stellarator. Presented at: Research School of Physics and Engineering, ANU.

* did not attend due to COVID19

Public Outreach Talks (10 total)

Aug. 2021	Building the Universe, Brick-by-brick. Presented at: Young Stars, ANU, Canberra.
May. 2021	Understanding The Big Bang. Presented at: Young Stars, ANU, Canberra.
Mar. 2021	The Secret Life of Cells. Presented at: Young Stars, ANU, Canberra.
Jan. 2021	Mission to Mars. Presented at: Young Stars, ANU, Canberra.
Jan. 2021	The Jiggling Universe. Presented at: SciScouts Space Squad, Canberra.
Nov. 2020	The Jiggling Universe. Presented at: Campbell Primary School STEM day, Canberra.
Oct. 2020	Thinking Like An Atom. Presented at: Young Stars, Canberra.
Sep. 2020	Simulating the Universe. Presented at: SciScouts Space Squad, Canberra.
Mar. 2020	Modelling Pandemics. Presented at: Young Stars, Canberra.
Feb. 2020	How do scientists test their ideas? Presented at: Young Stars, Canberra.

Teaching

Workshop lecturer

July - Aug. 2026	Plasma around Blackholes and Neutron Stars, SCEECS Summer School, Baja California, Mexico

Guest Lectures

Oct. 2026	University California, Santa Barbara: Guest lecture on MHD turbulence theory for a graduate level plasma class.
Sep. 2026	Dartmouth College: Guest lecture on MHD turbulence theory for a graduate level plasma class.
Oct. 2022	ASTR8002 (ANU): Guest lecture on MHD turbulence theory for a graduate level gas dynamics class.
Oct. 2020	ASTR8002 (ANU): Guest lecture on linear MHD waves for a graduate level gas dynamics class.

TA experience (**Click on the "Semester" to see teacher evaluation reports**)

2021	**Australian National University**, Canberra, Australia	
	ASTR2013: Foundations of Astrophysics	Semester Two
2018	**Queensland University of Technology**, Brisbane, Australia	
	PVB101: Physics of the Large	Semester Two
	MXB105: Calculus of One and Two Variables (wrote all assessment)	Semester Two
	MXB161: Computational Explorations	Semester One
	SEB113: Quantitative Methods in Science	Semester One & Two
	SEB104: Grand Challenges in Science	Semester One
	SEB115: Experimental Science	Semester One
2017	**Queensland University of Technology**, Brisbane, Australia	
	MXB105: Calculus of One and Two Variables	Semester Two
	PVB101: Physics of the Large (Lab Demonstrator)	Semester Two
	BVB204: Ecology	Semester Two
	SEB113: Quantitative Methods in Science	Semester One & Two
	SEB104: Grand Challenges in Science	Semester One
	SEB115: Experimental Science (Lab Demonstrator)	Semester One
	MXB161: Computational Explorations	Semester One
2016	**Queensland University of Technology**, Brisbane, Australia	
	PVB101: Physics of the Large (Lab Demonstrator)	Semester Two
	BVB202: Plant Biology (Lab Demonstrator)	Semester Two
	BVB224: Plant Diversity (Lab Demonstrator)	Semester Two
	SEB113: Quantitative Methods in Science	Semester One & Two
	SEB104: Grand Challenges in Science	Semester One
	SEB115: Experimental Science (Lab Demonstrator)	Semester One
2015	**Queensland University of Technology**, Brisbane, Australia	
	SEB113: Quantitative Methods in Science	Semester One

Publications (41 total)

- **First (and joint first) author:** 21 publications (17 refereed)
- **Total:** 41 publications
- **Citations:** 1,289 (12-Oct-2026)
- **h index:** 19 (12-Oct-2026)

First Author (and joint first) Refereed (17 total)

Beattie, **J. R.**, & Bhattacharjee, A. (2026). Local relaxation and scale-dependent alignment in compressible, magnetized turbulence. *Phys. Rev. Lett.* https://doi.org/10.1103/zp1g-rkjc

Beattie, **J. R.**, Sokolova, M., Negandhi, K., & Ripperda, B. (2026). Compressible navier-stokes flow in schrödinger-type variables. *Phys. Rev. Res.*, *8*, 033383. https://doi.org/10.1103/6gnr-vkx3

Beattie, **J. R.**, Federrath, C., Kriel, N., Hew, J. K. J., & Bhattacharjee, A. (2025). Taking control of compressible modes: bulk viscosity and the turbulent dynamo. *MNRAS, accepted*, Article arXiv:2312.03984, arXiv:2312.03984. https://doi.org/10.48550/arXiv.2312.03984

Beattie, **J. R.**, Kolborg, A. N., Ramirez-Ruiz, E., & Federrath, C. (2025). So Long Kolmogorov: The Forward and Backward Turbulence Cascades in a Supernovae-driven, Multiphase Interstellar Medium. *ApJ*, *994*(2), Article 193, 193. https://doi.org/10.3847/1538-4357/ae07cd

Beattie, **J. R.** (2025). Supernovae drive large-scale, incompressible turbulence through small-scale instabilities. *arXiv e-prints, Accepted in ApJL*, Article arXiv:2509.07354, arXiv:2509.07354. https://doi.org/10.48550/arXiv.2509.07354

Beattie, **J. R.**, Federrath, C., Klessen, R. S., Cielo, S., & Bhattacharjee, A. (2025). The spectrum of magnetized turbulence in the interstellar medium. *Nature Astronomy*. https://doi.org/10.1038/s41550-025-02551-5

Bandyopadhyay, R., **Beattie**, **J. R.**, & Bhattacharjee, A. (2025). Density fluctuation–mach number scaling in compressible, high plasma beta turbulence: In situ space observations and high-reynolds number simulations. *The Astrophysical Journal Letters*, *982*(2), L45. https://doi.org/10.3847/2041-8213/adbe3b

Beattie, **J. R.**, Federrath, C., Kriel, N., Mocz, P., & Seta, A. (2023). Growth or Decay - I: universality of the turbulent dynamo saturation. *MNRAS*, *524*(3), 3201–3214. https://doi.org/10.1093/mnras/stad1863

Birch, M., **Beattie**, **J. R.**, Bennet, F., Rattenbury, N., Copeland, M., Travouillon, T., Ferguson, K., Cater, J., & Sayat, M. (2023). Availability, outage, and capacity of spatially correlated, australasian free-space optical networks. *J. Opt. Commun. Netw.*, *15*(7), 415–430. https://doi.org/10.1364/JOCN.480805

Beattie, **J. R.**, Krumholz, M. R., Federrath, C., Sampson, M. L., & Crocker, R. M. (2022). Ion alfvén velocity fluctuations and implications for the diffusion of streaming cosmic rays. *Frontiers in Astronomy and Space Sciences*, *9*. https://doi.org/10.3389/fspas.2022.900900

Beattie, **J. R.**, Mocz, P., Federrath, C., & Klessen, R. S. (2022). The density distribution and physical origins of intermittency in supersonic, highly magnetised turbulence with diverse modes of driving. *MNRAS*. https://doi.org/10.1093/mnras/stac3005

Beattie, **J. R.**, Krumholz, M. R., Skalidis, R., Federrath, C., Seta, A., Crocker, R. M., Mocz, P., & Kriel, N. (2022). Energy balance and Alfvén Mach numbers in compressible magnetohydrodynamic turbulence with a large-scale magnetic field. *MNRAS*. https://doi.org/10.1093/mnras/stac2099

Beattie, **J. R.**, Mocz, P., Federrath, C., & Klessen, R. S. (2021). A multishock model for the density variance of anisotropic, highly magnetized, supersonic turbulence. *MNRAS*, *504*(3), 4354–4368. https://doi.org/10.1093/mnras/stab1037

Beattie, **J. R.**, Federrath, C., & Seta, A. (2020). Magnetic field fluctuations in anisotropic, supersonic turbulence. *MNRAS*, *498*(2), 1593–1608. https://doi.org/10.1093/mnras/staa2257

Beattie, **J. R.**, & Federrath, C. (2020). Filaments and striations: anisotropies in observed, supersonic, highly magnetized turbulent clouds. *MNRAS*, *492*(1), 668–685. https://doi.org/10.1093/mnras/stz3377

Beattie, **J. R.**, Federrath, C., Klessen, R. S., & Schneider, N. (2019). The relation between the turbulent Mach number and observed fractal dimensions of turbulent clouds. *MNRAS*, *488*(2), 2493–2502. https://doi.org/10.1093/mnras/stz1853

Beattie, **J. R.**, Federrath, C., & Klessen, R. S. (2019). The relation between the true and observed fractal dimensions of turbulent clouds. *MNRAS*, *487*(2), 2070–2081. https://doi.org/10.1093/mnras/stz1416

Second Author Refereed (8 total)

Connor, I., **Beattie**, **J. R.**, Kolborg, A. N., & Ramirez-Ruiz, E. (2025). Cascading from the winds to the disk: The universality of supernovae-driven turbulence in different galactic interstellar mediums. *Accepted in The Astrophysical Journal*. https://arxiv.org/abs/2509.01653

Kriel, N., **Beattie**, **J. R.**, Krumholz, M. R., Schober, J., & Armstrong, P. J. (2025). The growth of magnetic energy during the nonlinear phase of the subsonic and supersonic small-scale dynamo. *Accepted in PRD*, Article arXiv:2509.09949, arXiv:2509.09949.

Sampson, M. L., **Beattie**, **J. R.**, Teyssier, R., Kempski, P., Moseley, E. R., Commerçon, B., Dubois, Y., & Rosdahl, J. (2025). Cosmic ray and plasma coupling for isothermal supersonic turbulence in the magnetized interstellar medium. *Accepted in ApJ*, Article arXiv:2506.03768, arXiv:2506.03768. https://doi.org/10.48550/arXiv.2506.03768

Kriel, N., **Beattie**, **J. R.**, Federrath, C., Krumholz, M. R., & Hew, J. K. J. (2025). Fundamental MHD scales - II: the kinematic phase of the supersonic small-scale dynamo. *MNRAS*. https://doi.org/10.1093/mnras/staf188

Sampson, M. L., **Beattie**, **J. R.**, Krumholz, M. R., Crocker, R. M., Federrath, C., & Seta, A. (2023). Turbulent diffusion of streaming cosmic rays in compressible, partially ionized plasma. *MNRAS*, *519*(1), 1503–1525. https://doi.org/10.1093/mnras/stac3207

Kriel, N., **Beattie**, **J. R.**, Seta, A., & Federrath, C. (2022). Fundamental scales in the kinematic phase of the turbulent dynamo. *MNRAS*. https://doi.org/10.1093/mnras/stac969

McCool, C., **Beattie**, **J. R.**, Milford, M., Bakker J. D., J. L., Moore, & Firn, J. (2018). Automating analysis of vegetation with computer vision: Cover estimates and classification. *Ecology and Evolution*, *8*(12), 6005–6015. https://doi.org/10.1002/ece3.4135

McCool, C., **Beattie**, **J. R.**, Firn, J., Lehnert, C., Kulk, J., Bawden, O., Russell, R., & Perez, T. (2018). Efficacy of mechanical weeding tools: A study into alternative weed management strategies enabled by robotics. *IEEE Robotics and Automation Letters*, *3*(2), 1184–1190. https://doi.org/10.1109/LRA.2018.2794619

Major Contributions Refereed (5 total)

Publications which involved a significant amount of student mentorship or major contributions from JRB.

Grehan, M. P., Ghosal, T., **Beattie**, **J. R.**, Ripperda, B., Porth, O., & Bacchini, F. (2025). Comparison of magnetic diffusion and reconnection in ideal and resistive relativistic magnetohydrodynamics, ideal magnetodynamics, and resistive force-free electrodynamics. *Phys. Rev. D*, *112*(6), Article 063046, 063046. https://doi.org/10.1103/8xf2-x2nq

Risch, A. C., Page-Dumroese, D. S., Schweiger, A. K., **Beattie**, **J. R.**, Curran, M. P., Finér, L., Liu, Y., Schütz, M., Terry, T. A., Wang, W., & Jurgensen, M. F. (2022). Controls of initial wood decomposition on and in forest soils using standard material. *Frontiers in Forests and Global Change*, *5*, 829810. https://doi.org/10.3389/ffgc.2022.829810.

Constructed the principle data set, developed and ran parallelised hierarchical Bayesian mixed effects models and model selection methods.

Skalidis, R., Sternberg, J., **Beattie**, **J. R.**, Pavlidou, V., & Tassis, K. (2021). Why take the square root? An assessment of interstellar magnetic field strength estimation methods. *A&A*, *656*, Article A118, A118. https://doi.org/10.1051/0004-6361/202142045.

Ran all MHD turbulence simulations and contributed to the theoretical development of the coupling term energy model and drafting the manuscript.

Federrath, C., Klessen, R. S., Iapichino, L., & **Beattie**, **J. R.** (2021). The sonic scale of interstellar turbulence. *Nature Astronomy*, *5*, 365–371. https://doi.org/10.1038/s41550-020-01282-z.

Measured the sonic scale position from the second order structure functions and contributed to writing the manuscript.

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2024 CITA Fellow James Beattie was awarded close to 100-million-core-hours to explore the nature of turbulence and dynamos, *CITA*

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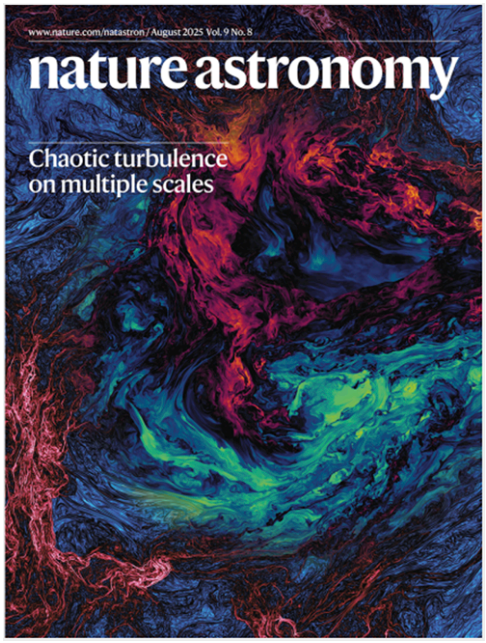
2024 The Beauty of Chaos, *New Scientist Magazine (in print and online)*

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2022 Unravelling magnetised turbulence in galaxies, *Lunations, Research Bytes*

2022 The Magic And Mystery Of Turbulence, *IFL Science*



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2021 The Role of Turbulence in the Birth of Stars, *University Heidelberg*

2021 Star-making motion, *COSMOS magazine*

2021 Study helps unlock secrets of star formation, *ANU Media*

2021 Stellar Simulation Reveals The Turbulent Nature of Star Birth, *Space Australia*

2021 The Need for (Sound) Speed, *Astrobite research highlight*

2019 Modelling Star Formation with a Supercomputer: Computational Astrophysics Research, *National Computing Infrastructure Australia*

2019 Feature article on turbulence depicted in Van Gogh's Starry Night in the Art's and Culture section of the American Physical Societies Magazine.

2018 QUT advertising photoshoot for the BSc and BMath degree on QUT's blackboard website.

2018 QUT media exposure, and photoshoot for our publication, Automating Analysis of Vegetation with Computer Vision: cover estimates.